

Self-Improving Neural Architectures

Networks that propose their own architecture modifications — meta-research: learning how to improve learning architectures automatically.

This is the frontier phase. Rather than designing a model, you design a system that designs models. An RL-trained architectural agent observes gradient statistics and loss surface curvature, then proposes modifications to the attention mechanism it is part of. The feedback loop — propose, evaluate, refine — is the core contribution.

CORE TASKS

- Build an architectural agent that proposes attention tweaks based on loss surface curvature and gradient metrics.
- Implement a feedback loop: modification → evaluation → refinement → new modification.
- Compare the agent's proposals to hand-designed modifications on held-out tasks.
- Constrain the search space to interpretable modifications (head count, kernel type, sparsity level, etc.).

MATHEMATICAL FORMULATION

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Agent input: s_t = [grad_L, H(L), rank(A), head_mi, step_t]
Agent output: a_t in {add_head, prune_head, swap_kernel, adjust_sparsity, ...}
Reward: r_t = L_before - L_after(a_t) + lambda · efficiency_gain(a_t)
Policy: pi(a|s) trained via PPO or REINFORCE over modification trajectories
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INSTRUMENTATION

- Track improvement metrics vs. hand-designed baselines across modification cycles.
- Log failed modifications and measure recovery efficiency and error patterns.
- Measure sample efficiency: how many trials before the agent outperforms random search?
- Analyze what the agent learns about architecture space — cluster successful proposal types.

DELIVERABLES

Type	Description
Code	Architectural agent with full modification → evaluation feedback loop

Code	Modification replay buffer and failure taxonomy logger
Analysis	Comparative study: agent-proposed vs. human-designed modifications over 50+ trials
Writeup	Full research report with open questions section pointing to future work

RESEARCH INSIGHT

This is beyond normal research — it is meta-research. You are not just training a model; you are training a system that improves the training process itself. This is the frontier of automated AI research.

Difficulty: Frontier | AutoML · RL for architecture · Meta-research · Self-modification